A survey of encrypted traffic classification: datasets, representation, approaches and future thinking

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***Abstract*—Traffic classification technologies have evolved as a key component for modern network protection and management systems. However, encryption algorithms have been adopted in traffic transmission to enhance communication security and privacy protection. Traditional traffic classification methods that rely on valuable information from payloads gradually become ineffective. To this end, researchers have conducted a lot on encrypted traffic classification. In this paper, we present a comprehensive survey on recent achievements in traditional-machine-learning (TML) or deep-learning (DL) powered encrypted traffic analysis. To begin with, we review the literature in this area and summarize the analysis goals. Secondly, we abstract the overall framework of encrypted traffic classification, including traffic collection, representation, analysis methods, and performance evaluation. Then a comprehensive review of existing research was conducted from these four aspects. Lastly, we conclude with challenges and directions for future research in encrypted traffic classification.**

Keywords—encrypted traffic classification, traffic analysis, machine learning, deep learning

I. Introduction

Encrypted network traffic is encrypted by encryption protocols during transmission, which makes it difficult to be eavesdropped or tampered. It can be used to protect user privacy and data security, and also prevents hackers or malicious attackers from stealing and tampering with transmitted data. There are many encryption protocols used to encrypt network traffic, ranging from the network layer, transport layer to the application layer, such as IPSec, SSL/TLS, QUIC, SSH, Tor, etc.

According to the statistics [1], by the end of January 2024, more than 96% of the traffic on the Google website is in HTTPS, and the proportion of web pages loaded through HTTPS on both Windows and Android platforms exceeds 96%. The encryption usage rate of domestic websites also shows an upward trend and has occupied a high proportion. Thus, how to effectively identify massive amounts of various encrypted network traffic is a huge challenge. Generally speaking, encrypted traffic identification can be divided into three types according to different business needs,: (1) Encrypted traffic and non-encrypted; (2) Identification of different encrypted traffic types, such as chat, video streaming, email, etc.; (3) Specific APP identification, such as TikTok, Alipay, WeChat, QQ, etc.

At present, a large number of researchers have reviewed traffic identification. Compared with the existing surveys, this article does not target specific application identification, nor does it target specific usage scenarios. We only comprehensively summarize, analyze and prospect encrypted traffic identification methods, including TML and deep DL methods. The main contributions of this survey are summarized as follows:

- Abstract the overall workflow of encrypted traffic analysis from a great number of current approaches, which presents an overview that helps readers grasp the general process of encrypted traffic analysis.
- The first to make a comprehensive review of existing research conducted from these four aspects, which guide readers at different research stages.
- Provide further thinking into the research challenges and directions on encrypted traffic analysis, which provides readers with possible directions for developing innovative solutions.

The reminder of this paper is organized as follows. Section II gives a brief description of the workflow of encrypted traffic analysis. The next four sections introduce the traffic collection (Section III), representation (Section IV), analysis methods (Section V), and performance evaluation (Section VI), respectively. The article ends with conclusion in Section VII.

II. Overall Workflow of Encrypted Traffic Analysis

We summarize our general workflow for encrypted traffic analysis with four components, namely traffic collection, traffic representation, traffic analysis method, and performance evaluation.

Traffic collection is an important step in encrypted traffic analysis, as a data set is the foundation of experiments. The experimental data of most research are based on standard datasets published previously (such as ISCX VPN, ISCX Tor, etc.) or self-constructed datasets. And also, some scholars generate training data by enhancement methods. All in all, data collection often acts as the starting point of encrypted traffic analysis, which will be detailed in Section III.

Traffic representation is essential for encrypted traffic classification. For different application scenarios and analysis methods. Better representations can better mine the semantic information and the differences between different encrypted traffic. Most existing methods construct representations of encrypted traffic from the packet, level, and session-level. And the feature types are not limited to content, statistical features, etc. Details will be introduced in Section IV.

Traffic analysis methods are chosen based on traffic representation and different tasks. The current mainstream encrypted traffic identification methods include traditional-machine-learning and deep-learning based methods, which will introduced in Section V.

Performance evaluation is used to confirm that the analysis methods are suitable for their intended use. The accuracy-related indicators are mainly used in encrypted traffic analysis evaluation. Some others introduced compatibility, robustness, and completeness to evaluate encrypted traffic identification technology more comprehensively and continuously (Section VI).

III. Traffic Collection

A. Datasets from existing work

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In Table I, there are a summary of datasets used in recent existing work. However, there are many problems with public datasets, such as (1) No dataset can contain all types of traffic due to the huge numbers of traffic types and frequent updates of applications. (2) It is very hard, time-consuming and costly to cover all the network scenarios.

TABLE I. THE SUMMARY OF DATASETS FORM EXISTIONG WORK

Dataset	*Year*	*Description*	*Link*
CTU-13	2011	(1) real botnet traffic mixed with normal traffic and background traffic (2) consists in thirteen captures (called scenarios) of different botnet samples	https://www.stratosphereips.org/datasets-ctu13
ISCX 2012	2012	(1) HTTP, SMTP, SSH, IMAP, POP3, and FTP	https://www.unb.ca/cic/datasets/ids.html
USTC-TF 2016	2016	(1) 10 Bengin (BitTorrent, Facetime, FTP, Gmail, MySQL, Outlook, Skype, SMB, Weibo, WorldOfWarcraft), 10 Malware (Cridex, Geodo, Htbot, Miuref, Neris, Nsis-ay, Shifu, Tinba, Virut, Zeus)	https://www.kaggle.com/datasets/randasrour/ustctfc2016
ISCX VPN 2016	2016	(1) 14 traffic categories: VOIP, VPN-VOIP, P2P, VPN-P2P, etc (2) 28GB of data	https://www.unb.ca/cic/datasets/vpn.html
ISCX Tor 2016	2016	(1) 7 traffic categories: Browing, Email, Chat, Streaming, FTP, VoIP, P2P (2) 22GB of data	https://www.unb.ca/cic/datasets/tor.html
CIC-IDS 2017	2017	(1) provided all common available protocols, such as HTTP, HTTPS, FTP, SSH and email protocol (2) included the most common attacks, such as Web based, Brute force, DoS, DDoS, Infiltration, Heart-bleed, Bot and Scan covered	https://www.unb.ca/cic/datasets/ids-2017.html
IMTD 17	2017	(1) included 12 different mobile applications, such as Alipay, Baidu, Bilibili, CNTV, JD, Kugou, QQ, QQ Mail, QQ Music, Taobao, WeChat (2) 61000 samples	https://github.com/ld258166011/IMTD17
SJTU-AN21	2021	(1) Eepsites, IRC, Snark, and Video are the four main anonymity services in the I2P network (2) BitTorrent, Chat, FTP, Streaming, and Browsing are the five main anonymity services in the Tor network (3) 36193 samples	https://github.com/iZRJ/The-SJTU-AN21-Dataset

B. *Datasets from self-buliding tools*

When existing datasets cannot meet the needs of detection tasks or in a new network scenario, some work collected raw data from ISP's network or research lab to create their own dataset. (1) Using online tools such as JOY and Bro (Zeek) to create datasets. (2) Under different network environments (5G, 4G, 3G, etc.), replaying the existing data to obtain the new data constructed under the different network scenario.

IV. TRAFFIC REPRESENTATION

This section mainly summarizes existing methods from the different levels and types of traffic representation.

A. *The levels of traffic representation*

Traffic is transmitted in bit streams, consisting of packets, flows, sessions, etc. Based on these basic compositions, the feature representation of existing encrypted network traffic identification methods can be mainly divided into the following three levels: packet, flow, and session level.

a) Packet-level: Representation focuses on features related to the packet-self and its arrival process, including packet size, protocol, raw flow, etc. Boukhtouta et al.[2] extracted 29 packet-level features from the data link layer, network layer and transport layer to represent each packet. Liu et al.[3] added packet protocol features to the packet size features and achieved malicious traffic detection using only the features from the first 8 packets (7 package features, 8 protocol features, 8 certificate features). Relatively speaking, Luo et al.[4] used sliding window technology to explore which bytes in packets contribute the most to encrypted network traffic classification.

Most of the above-mentioned approaches are based on the raw bytes or statistical features of packets as model input. With the development of deep learning, many researchers use deep learning methods to extract higher-dimensional features from the packet stream.[5] first applied representation learning to malicious traffic identification, which extracted the first 784 bytes of the raw byte stream of packet and used CNN to automatically learn the flow features. [7] regarded the header information of packets as a sentence (every 2 bytes as a word), and constructed packet embedding by word embedding word2vec[8]. In addition, packet-level features also include features of the packet arrival process, such as packet arrival time, arrival time interval distribution, etc. Gil et al.[9] studied a VPN-encrypted traffic detection method based on packet time-related features. [10] only combines the two most basic features of packet size and packet arrival interval time to achieve encrypted traffic classification.

b) Flow-level: The flow-level representation focuses on network flow-related features and their arrival processes, including flow duration, counts of flow bytes, communication mode, flow sequence relationship, etc.

Qin et al. [11] used a bidirectional flow model to aggregate traffic packets into bidirectional flows, thereby capturing the interactive behavior between different terminals. Similarly, Lokoč et al. [12] divided information such as the flow bytes of upstream and downstream, link duration, and session request interval duration into several intervals, and constructed a four-tuple vector of 114 features. Based on the work of Lokoč et al.[12], Kohout et al.[13] improved the expression of the four-tuple vector for flow communication patterns, and constructed an intra-flow or inter-flow representation model for malicious traffic identification.

Like the raw traffic representation at the packet level, some researchers have proposed the raw flow representation for the deep learning model. Marín et al.[14] compared the representations of packet-level (first 1024 bytes) and flow-level (first 2 packets, 100 bytes of each packet). The experimental results show that the flow-level used as the input of the deep learning model obtains better detection performance. Liu et al.[15] deeply mine the latent sequential features of flows by adopting a multi-layer encoder-decoder structure. Since network data is essentially a non-Euclidean distance space and contains rich temporal and spatial relationships, Huoh et al.[16] comprehensively considered the time and space features of flow and proposed a graph neural network (GNN) based encrypted traffic identification method. Compared with previous supervised encrypted traffic identification methods, Fu et al.[17] proposed an unsupervised malicious traffic detection system that can detect unknown patterns through the flow-interaction graph.

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c) Session-level: The session-level representation focuses on the relevant features of sessions and their arrival processes. A session may contain multiple flows, and a flow consists of one or more packets. Therefore, the features extracted from the packet-level and flow-level are also suitable for session-level representation, such as the number of flows, the average duration of flows, etc., can be converted into session-level features. Cui et al.[19] used a capsule neural network to construct an encrypted traffic classification model SPCaps based on session packets. Hong et al.[20] regarded sessions as nodes and constructed an attribute KNN graph for discovering encrypted malicious traffic.

B. *The types of traffic representation*

Different types of traffic representation correspond to inputs to different types of classification models. Specifically, the types of feature representation can be divided into: **content features, statistical features, and raw flow features**. The first two are generally used as inputs for traditional-machine-learning models. The raw flow features (such as sequences[21] or graphs[22]) can be used as inputs for deep-learning models.

a) Content features: In the early Internet days, traffic was non-encryption or partial-encryption. Content features could be used to identify encrypted network traffic, such as port number, payload information, etc. However, with the rapid development of the Internet, it is increasingly unfeasible to rely on fixed port numbers to identify traffic. Madhukar et al.[23] found that the identification method based on port numbers cannot identify 30% to 70% of the traffic, and the stability of the method is relatively poor. Payload information can be obtained by analyzing the payload of packets or flows[24]. However, this type of method requires parsing the content of the packet, which easily violates personal data privacy. At the same time, when dealing with private protocols or encrypted protocols, it is difficult to obtain due to application layer data encryption and must be updated simultaneously when the protocol changes.

b) Statistical features: Statistical features are extracted from network traffic through abstraction or calculation, which can represent its basic attributes and behavioral paradigms. Commonly used statistical dimensions include expectation, deviation, average, minimum, maximum, median, accumulation, etc. By combining these statistical dimensions with traffic, a large number of statistical features are generated, such as the average cumulative packet size, average inter-arrival time, etc.

TABLE II. THE COMMON STATISTICAL FEATURES

Level	*Feature*	*Accompany-feature*
Packet	port number	source, destination
	Protocol number	/
	number	source to destination, destination to source, flag bits (URG, ACK, PSH, RST, FIN, SYN, CWR, ECE)
	size	source to destination (max, min, average, standard deviation), destination to source (max, minimum, average, standard deviation), bidirectional flow (max, minimum, average, standard deviation), size of data packets transmitted per second, size of data packets transmitted per second in bytes
	arrival time	source to destination (maximum, minimum, average, standard deviation), destination to source (maximum, minimum, average, standard deviation), bidirectional flow (maximum, minimum, average, standard deviation)
Flow	duration	/
	sending rate	sending rate (in bytes)
	length	flow (total, maximum, minimum, average, standard deviation), upstream (total, maximum, minimum, average, standard deviation), downstream (total, maximum, minimum, average, standard deviation)
	number	total number, total number of upstream, total number of downstream
Session	statistical features of packet and flow are suitable for sessions	

c) Raw flow features: Content and statistical features require manual extraction for encrypted traffic classification. It relies on manual experience, time-consuming and labor-intensive, and may not necessarily achieve the expected results. Therefore, researchers have begun to extract features automatically with deep-learning algorithms from raw flow. Thereby the burden of feature extraction is shifted to the model. Some studies convert traffic into images, sequences and graphs to use CNN[14][19], RNN[15], GNN[16][20] to represent encrypted traffic. The algorithms of deep feature representation will be highlighted in Section V.

V. TRAFFIC ANALYSIS METHODS

This section outlines typical encrypted traffic identification methods based on traditional machine learning (TML) and deep learning (DL).

A. *TML-based methods*

The TML-based methods extract features from encrypted traffic and train ML models to classify traffic. They have achieved good performance in the three tasks mentioned above of encrypted traffic identification[34]. This sub-section mainly reviews models in encrypted traffic classification from three aspects: supervised, unsupervised and semi-supervised.

a) Supervised methods: Supervised methods are trained with labeled traffic data for prediction. The common methods include Naïve Bayes, Markov Chain, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), etc.

Naïve Bayes. Bayesian algorithm calculates the similarity of each feature of flows to identify encrypted or non-encrypted traffic, as well as other identification tasks. [26] learn the random features of Skype in average packet arrival time and packet length via Naïve Bayes, and the identification false positive rate is almost zero. [27] and [39] respectively classify the protocols and specific applications of SSL/TLS by Bayesian algorithm. The former [27] mainly uses packet-level features to achieve about 95% F1 (such as HTTPS, Tor, UPDATA, OSCAR, etc.). The flow-level features are added in the latter [39], which uses a total of 67 statistical features to classify 44 specific applications (including Twitter, Facebook, Amazon, etc.). The Bayesian algorithm is also adopted in [36] with packet and flow-level features (about 49) to realize the identification of HTTP, FTP, SSH, and SMTP protocols, with an accuracy of almost 100%. In addition to packet and flow-level features, session-level features are added in AlSabah et al. [38], in which a Bayesian algorithm is adopted to learn time interval, session activity and session volume to achieve real-time identification of different services in Tor.

Markov Chain. Most existing research captures the changes in encryption policy and certificate packet length in

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traffic via Markov chains to realize encrypted traffic classification. Korczyński et al. [25] found that the message types of SSL/TLS always changed during data transmission in Twitter and PayPal, thus the first-order Markov chain was introduced to capture changes for fingerprint construction. On this basis, [40] added the certificate package length information in building the fingerprint based on the second-order Markov chain, and the recognition accuracy increased by about 30%. Similarly, [41][42] further improved the accuracy of encrypted traffic identification based on the Markov chain by involving packet size and length sequence.

KNN. The KNN algorithm is easy to implement and can handle a large number of traffic categories. [29] established Web FingerPrints (WFP) with packet-level statistical features and achieved a 91% identification rate based on the KNN in a closed environment. [45] only uses the statistical features of the first 100 packets during the process of web loading, and implements fine-grained early web recognition based on KNN. Compared to the local statistical features of web traffic in [29][45], Gong et al. [43] adopted Dynamic Time Warping to implement a KNN-based algorithm for the complete time series of web traffic, which also achieved 80% accuracy in an open environment. Based on multi-type flow-level features, [44] constructed a fine-grained classification method based on the hierarchical KNN, and achieved good performance on large-scale real video traffic.

SVM. SVM has also been widely used in the field of encrypted traffic. [28] proposed SVM-based Web traffic identification algorithm, which is trained by flow-level WFP. [47] further proposed new features to describe the similarity between traffic for SVM, the accuracy in Tor in a closed environment was as high as 91%. Yuan et al. [46]adopted SVM to learn flow-level statistical features to classify encrypted services, accuracy reached 97.17%. Similarly, [50][51][52] extract a variety of packet-level and flow-level statistical features and propose or improve SVM-based algorithms in the field of encrypted traffic classification. In addition, the traditional SVM has high training complexity and large memory/CPU computing overhead. To this end, Sun et al. [48] proposed incremental SVM to reduce the costs of memory and CPU. The model utilizes new training datasets and previous-generated support vectors to achieve rapid updates in training. Megantara et al. [49] combined the wrapping method in the linear SVM algorithm with the variance analysis method. The method can distinguish features used or not, which improves the training speed of SVM-based classification and shortens detection time.

DT. In encrypted traffic classification, DT can involve the content or statistical features of packet-level, flow-level, and session-level as model inputs. DT automatically selects the feature combination during the training process and achieves encrypted traffic classification. [35] defines 22 flow-level features for encrypted application classification and verifies the computing performance (modeling time and Classification speed) of 4 different TML-base methods. The accuracy of each model is similar, but the C4.5 has a faster classification speed. Similarly, [36][37] evaluated 5 (AdaBoost, SVM, Naïve Bayes, RIPPER and C4.5) and 6 (LR, SVM, Bayesian, KNN, RF, GBT) TML-based methods respectively in identifying SSH and Skype, VPN or non-VPN. In which DT performed better. [53][54] compared multiple TML-based algorithms in corresponding scenarios and reached the same conclusion. Relatively speaking, DT has faster calculation speed and higher classification accuracy. Apart from previous research, Tong et al. [55] proposed a real-time classification method based on C4.5, which achieved a real-time classification accuracy of up to 97.92%.

RF. RF is a classifier that contains multiple DTs, and the category of its output is determined by the mode of individual trees. [30] adopted RF to extract page fingerprints and trained by traditional packet-level statistical features, which achieved an accuracy of 91% in a closed environment. [31][32][56][57] mainly introduced RF to achieve Quality of Experience (QoE) estimation for encrypted video traffic. Apart from the non-real-time analysis, [57] subdivided the session in the YouTube video into 1s streams, and implemented real-time analysis based on RF. [58][59] achieved the classification of encrypted traffic by improving the feature selection process or optimizing the RF algorithm.

b) Unsupervised methods: Compared with supervised methods, unsupervised methods cluster unlabeled data in the training set based on the similarity of encrypted traffic. The common unsupervised learning method is K-means. Erman et al. [60] first introduced K-means in encrypted traffic classification and achieved well performance. [63] evaluated two different unsupervised algorithms K-means and the Maximum Expectation (ME) in the three tasks of encrypted traffic, in which K-means achieved higher accuracy than EM.

Additionally, researchers have carried out research in terms of feature construction of unsupervised methods and model optimization. [62] mainly considered the port number and payload length of the packet, and proposed a P2P traffic identification method based on the K-means. Distinguishing from [62], [33][61] implemented a real-time classification method for encrypted traffic based on cluster analysis. Liu et al. [64] and Xiao et al. [65] started from the perspective of unsupervised algorithm model optimization.

c) Semi-supervised methods: The methods are a combination of supervised and unsupervised, aiming to utilize a combination of labeled and unlabeled data for model training. It is more flexible in sample data annotation and can make full use of the information of unlabeled data to improve model performance and generalization. Zhang et al. [66] proposed a semi-supervised traffic classification method based on nearest-neighbor propagation learning. To overcome the time-varying changes in behavioral features, [67] proposed a reliable semi-supervised intrusion detection model. In addition, [68] proposed an improved semi-supervised traffic classification framework based on BIRCH, which was compared with supervised and semi-supervised algorithms with higher classification accuracy.

B. DL-based methods

TML-based methods extract features manually, and the selection of features will directly affect the classification performance. It requires a certain amount of manpower and material investment, which limits the wide applicability of TML-based methods. On the contrary, due to its powerful feature representation, DL-based methods have successfully solved several problems such as poor adaptability and waste of manpower. In 2015, Wang et al. [69] first tried to use deep learning (Multi-Layer Perceptron) for traffic classification, which reached a precision and recall above 90%. After that, researchers began to explore convolutional neural networks (CNNs), recurrent neural networks (RNNs), graph neural networks (GNNs), etc. in encrypted traffic classification. This sub-section will provide an overview from the perspective of

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the application of different deep learning models.

a) CNNs-based methods: CNNs consist of one or more convolutional layers and a fully connected layer, as well as associated weights and pooling layers. It can adapt high-dimensional vectors by convolutional layers and use convolution and pooling to reduce model parameters. [6] only extracted the first 784 bytes of the raw flow and adopted 1d-CNN to automatically classify. Rezaei et al. [78] proposed a semi-supervised encrypted traffic identification model based on 1d-CNN, which was trained on large unlabeled and small labeled datasets respectively. In addition, Zhang et al. [70] proposed a 3d-CNN-based encrypted traffic classification method, which outperformed 1d-CNN-based and 2d-CNN-based. [18] extracted the first 5000 packets, and adopted CNN to learn features of packet length to realize identification, achieving an accuracy of 98.3% in a closed environment. To reduce the length of packets in CNN-based models, Salman et al. [71] proposed a hierarchical CNN-based encrypted traffic identification method with two steps, which achieved a recognition accuracy of 95%. Apart from regarding flows as input to CNN, Chen et al. [72] and Shapira et al. [73] convert encrypted flows into images.

b) RNNs-based methods: RNNs can effectively handle sequence problems and have a certain memory for previous data sequences. Vu et al. [79] introduced LSTM to learn the time dependence of traffic to identify services. The method adopted the arrival time features of the first 5 packets as the initial input of the LSTM model according to the arrival time, which got 98% F1 in service identification in VPN traffic. [80] and [81] were both to address the problem of data imbalance.

To solve the poor parallelism and low operation efficiency in RNNs-based methods, Zhang et al. [82] proposed an encrypted traffic classification method based on the multi-layer bidirectional simple recurrent unit (SRU) and attention mechanism. [83] introduced a multi-modal deep learning framework (PEAN) based on LSTM, which demonstrated the effectiveness and efficiency of PEAN through experiments.

c) GNNs-based methods: With the widespread of CNNs and RNNs in encrypted traffic classification, GNNs have also begun to be involved in this field to capture spatial features[16]. Regarding the graph construction of encrypted traffic, researchers started with two aspects packets and flows. [84] and [85] converted packets in encrypted traffic into graphs. In terms of graph construction of flows, Cai et al. [86] adopted Markov chains to mine the topological information hidden in the flows to build a flow graph, and proposes an encrypted traffic classification method (MEMG) based on Markov Chain and GNN. In addition, researchers also conducted on the graph neural network structure itself, such as adding the attention mechanism to GNN [87], using k-WL to improve the generalization ability of the GNNs-based model [88] or constructing a temporal graph neural network [89], etc. In view of the features of ultra-high speed and large connections of 6G, some researchers have exploratorily proposed an encrypted malicious traffic detection method (ET-RSGAT) based on GNNs [90].

d) Hybrid Neural Networks (HNNs) based methods: Encrypted traffic classification methods with a single neural network will come up with the problems of incomplete features or a single structure. Thus, HNNs-based have been proposed, which are based on convolutional and recurrent neural networks and their variants.

[91][92] combined CNN and LSTM with the serial or parallel network structure. In which CNN and LSTM learned the spatial structure and the time series features of packets or flows separately. The combines exceed the single neural network models in various types of tasks. In addition, variant networks of RNNs (such as GRU, Bi-GRU, etc.) and CNNs (such as Inception-CNN, Capsnet, ResNet, etc.) achieved good performance in encrypted traffic classification. [93][94] introduced CNN and GRU to learn the spatial and sequence features of network flow respectively. Based on CNN and GRU, [97] added the attention mechanism to highlight important packets in the session. [95][96] used Bi-GRU to obtain bidirectional sequence information in flows.

In the above-mentioned, the sequence features are mined by variant RNNs-based methods. The introduction of variant CNNs-based methods can make the classification model more sensitive to local or global spatial features of encrypted traffic. Lu et al. [98] proposed a neural network (ICLSTM) based on parallel Inception-CNN and LSTM, which introduced the inception module in CNN to capture the local spatial features and used the LSTM to extract sequence features. To improve the ability of hierarchical representation, [99] introduced a deep neural network joint CapsNet and Bi-LSTM. In addition, to slowing the gradient disappearance caused by the superposition of network structures, [101] and [101] combined the Inception-CNN and ResNet, which greatly reduced the model complexity.

VI. Performance Evaluation

Generally speaking, accuracy is mainly adopted to evaluate the classification tasks, which is relatively single and not comprehensive. To meet the ever-increasing requirements for traffic analysis, researchers have introduced indicators such as completeness, robustness, portability, and so on.

A. Accuracy

Accuracy reflects the ability of traffic identification technology to identify encrypted network traffic, including precision, recall, F1 and accuracy. Taking the second task of encrypted traffic classification as an example, N is the number of traffic samples, m is the number of application types, n_{ij} represents the number of samples with the actual type i marked as type j. In which j represents the number of samples except i. As shown in Table III.

TABLE III. The Typical Encrypted Network Protocols

	Predicition type-i	***Prediction non-type-i***
Belong-type-i	TP_i	FN_i
Belong-non-type-i	FP_i	TN_i

According to the above, the precision, recall, F1 and accuracy are defined as follows,

$$precision = \frac{TP_i}{TP_i + FP_i}$$

$$recall = \frac{TP_i}{TP_i + FN_i}$$

$$F1 = \frac{2\,presion * recall}{presion + recall}$$

$$accuracy = \frac{\sum_{i=1}^{m}(TP_i + TN_i)}{\sum_{i=1}^{m}(TP_i + TN_i + FP_i + FN_i)}$$

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B. Completeness

Completeness is the coverage of the encrypted traffic classification method. Completeness is the ratio of the sample identified as i to the actual sample of type i, which is equivalent to the ratio of precision and recall. The value range may exceed 1. Defined as follows,

$$completeness = \frac{recall}{precision}$$

C. Unrecgnized

Unrecognized is the ability of the classification method to identify unknown traffic, which is the ratio of traffic that does not belong to known with the total traffic. F_{total} is the total number of bytes or flows of the traffic, and F_{known} is the number of bytes or flows of the identified traffic.

$$unrecognized = \frac{F_{total} - F_{know}}{F_{total}}$$

D. Portability

Portability is the ability of traffic classification methods to be used in different network environments. acc_j is the accuracy in environment j, and $\overline{acc}$ is the average accuracy.

$$portability = \sqrt{\frac{\sum_{j=1}^{m}(acc_j - \overline{acc})^2}{m}}$$

E. Robustness

Robustness is the ability of traffic classification methods to maintain a high identification rate over a long period. acc_k is the accuracy of period k, and acc_0 is the initial accuracy.

$$robustness = \sqrt{\frac{\sum_{k=1}^{r}(acc_0 - acc_k)^2}{r}}$$

VII. Challenges and Future Thinking

A. Challenges

a) Insufficient feature information: The advent of the full encryption era has resulted in insufficient plaintext information. The payload commonly used cannot be adopted to classify encrypted traffic. It is necessary to mine hidden features to bring incremental information for identification.

b) Target confrontation ability enhanced: As the confrontation between network security attacks and defenses intensifies, targets will continue to be iterated, optimized, upgraded and even changed. The features of encrypted traffic will also change accordingly, which will cause the performance of previously trained models to gradually decline.

c) Lcak of labeled samples: The TML-based and DL-based methods rely on a large number of labeled samples. It not only requires a lot of manpower but may also risk infringing user privacy. Moreover, new traffic in the early emergence is all small or zero samples.

d) Open enviroment identification: The number of applications is over one million in the real world. The theoretical basis of most algorithms is to input all data into the model for training. However, this is unrealistic.

B. Future Thinking

a) Construct the dataset in a real network environment: Simulate the real network environment to build an attack and defense confrontation scenario, which is close to the real network environment. And construct the dataset in this environment as much as possible for the evaluation of models, so that the performance impact of the model after it goes online is minimized.

b) Exploring patterns in encrypted traffic based on AI: AI is widely used in the fields of image and speech recognition. However, AI is still in its infancy in traffic recognition. There are still great challenges and many techniques that need to be broken through. AI-based encrypted traffic detection will also be a long-term research topic.

c) Not give up traditional methods: Due to the limitations of cryptography, the development of encryption protocols is also gradual. Thus, some statistical features are still effective. We need to conduct an in-depth understanding and analysis of traffic and propose targeted solutions.

d) Construct a hierarchical detection framework: The composition of traffic in the network is very complex, and the classification of encrypted traffic has never been solved by a single model. In practical use, a layered classification method can be built to solve different problems at different levels.

C. Conclusion

This paper surveys significant studies in the field of multi-method encrypted traffic analysis. In particular, we summarize the research in traffic collection, traffic representation, traffic analysis method, and performance evaluation. Lastly, We provide further thinking into the research challenges and directions on encrypted traffic analysis, which provides readers with possible directions for developing innovative solutions. In conclusion, determining how to properly use encrypted traffic analysis is a problem worthy of consideration. With the rapid development of the network and related technologies, encrypted traffic analysis will be a vibrant field and continue to flourish.

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